RELATIVE AMPLENESS OF AN INVERTIBLE SHEAF IS INSENSITIVE TO THICKENING THE SOURCE AND THE BASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Relative ampleness is checked on affine opens of the base. Morphisms, Definition labelled morphisms-definition-relatively-ample; invoked at lines 321-323. Reduces the relative statement to affine bases.
- Affine opens of S correspond to affine opens of the thickening S'. Same chapter, Lemma labelled lemma-thickening-affine-scheme; invoked at lines 324-326. Transports the affine-open reduction across the thickening and identifies the restricted situation with (X subset X') over (S subset S').
- Ampleness is insensitive to passing from a thickening to its reduction. Limits, Lemma labelled limits-lemma-ample-on-reduction; invoked at line 331. Supplies the affine case of the equivalence.

Source reference. Original lemma 06AD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 157–157. Navigation pointer for \ref{lemma-thickening-affine-scheme} used inside the counted span (line 324); the argument that consumes it is retained in the span.

Source reference. Original lemma 09MW: sources/stacks-chow/tags.txt, source lines 12194–12194. Navigation pointer for \ref{limits-lemma-ample-on-reduction} used inside the counted span (line 331); the argument that consumes it is retained in the span.

Source reference. Original lemma 01VH: sources/stacks-chow/tags.txt, source lines 2446–2446. Navigation pointer for \ref{morphisms-definition-relatively-ample} used inside the counted span (line 323); the argument that consumes it is retained in the span.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(f, f') : (X \subset X') \to (S \subset S')$ be a morphism of thickenings. Let $\mathcal{L}'$ be an invertible sheaf on X' and denote $\mathcal{L}$ the restriction to X. Then $\mathcal{L}'$ is f'-ample if and only if $\mathcal{L}$ is f-ample.*

Proof. Recall that being relatively ample is a condition for each affine open in the base, see Morphisms, Definition 01VH. By Lemma 06AD there is a 1-to-1 correspondence between affine opens of S and S'. Thus we may assume S and S' are affine and we reduce to proving that $\mathcal{L}'$ is ample if and only if $\mathcal{L}$ is ample. This is Limits, Lemma 09MW. □